GRADUATION PROJECT

MODEL2PRODUCT AI

Requirements Gathering & Analysis

AI Tech Lead Mentor & Project Delivery Simulator

22	4	10
FUNCTIONAL REQS	TEAM MEMBERS	MVP CHECKS

Stakeholder needs, traceable specifications, verifiable acceptance criteria and a practical validation plan.

DOCUMENT TYPE

Requirements analysis | Working draft

BASELINE

Proposal + 4-member plan

Prepared 16 October 2026

Owner / supervisor approval: pending

02 / DOCUMENT CONTROL

What is confirmed - and what is not

A proposal-based requirements baseline, not a claim that stakeholder interviews have already happened.

DOCUMENT PROFILE

CONTROL ITEM	DETAIL
Document status	Draft for stakeholder validation; not an approved SRS.
Project	Model2Product AI - AI Tech Lead Mentor & Project Delivery Simulator.
Delivery team	4 members; roles assigned by the team during kickoff.
Time assumption	No fixed week count or project duration is assumed.
Primary source	Model2Product AI Graduation Project Proposal (August 2026).
Secondary source	Revised Project Planning & Management (4-member team).

EVIDENCE CLASSIFICATION

A / PROPOSAL-DEFINED

Explicit functions, target users, stage-gated workflow and minimum demo acceptance.

B / ANALYST-DERIVED

User stories, test wording, priority tags and workflows translated from source requirements.

C / VALIDATION NEEDED

Open decisions about permissions, scoring rules, supported files, quotas and exact quality targets.

D / OPTIONAL REFERENCE

External engineering standards used as design aids, not proof of implementation or compliance.

APPROVAL RULE

No requirement should be treated as final until the product owner / supervisor reviews priorities, scope and test criteria.

03 / ELICITATION APPROACH

Requirements gathering method

Separate completed desk research from planned interviews and validation activities.

ACTUAL EVIDENCE AVAILABLE TODAY

COMPLETED / DOCUMENT ANALYSIS

Source proposal reviewed for stakeholders, 22 functional requirements, 10 non-functional categories, stage-gated behavior and 10 minimum acceptance checks.

RECOMMENDED NEXT ELICITATION ACTIVITIES

01 **Learner interviews**

2-4 students / junior AI engineers

Pain points, skills, project workflow, explanation depth.

02 **Instructor interview**

Supervisor / teaching assistant

Rubrics, learner evidence, auditability, class use.

03 **Scenario walkthrough**

One end-to-end sample AI project

Test discover - decide - approve - implement sequence.

04 **NFR + safety workshop**

4-member implementation team

Authentication, isolation, latency, resources, security.

05 **Review & prioritization**

Team + approving supervisor

Confirm Must / Should / Could and sign off backlog.

ELICITATION OUTPUTS

Interview notes, validated personas, prioritized backlog, clarified acceptance criteria, decision log and stakeholder sign-off.

04 / STAKEHOLDERS

Who has a requirement?

Users, buyers and reviewers have different objectives and must not be treated as one persona.

STAKEHOLDER NEEDS MATRIX

STAKEHOLDER	POSITION	KEY NEED
Learner / junior AI engineer	Primary user	Guided steps, credible options, hints, skills feedback.
Instructor / mentor	Reviewer	Progress evidence, weak areas, decision rationale, AI contribution.
University / academy	Potential buyer	Learner completion, quality evidence and simple rollout.
Team / project lead	Delivery stakeholder	Task breakdown, scope, dependencies, approvals.
Evaluation committee	Acceptance stakeholder	Working demo, traceability, tests, defensible artifacts.

INTERVIEW PROMPTS TO VALIDATE

LEARNER QUESTIONS

Where do you get stuck after training a model? What decisions would you struggle to justify? When is code generation helpful?

INSTRUCTOR QUESTIONS

What is sufficient evidence of understanding? Which steps must be mandatory? What needs to be visible in the dashboard?

INSTITUTION QUESTIONS

How will cohorts be managed? What user information may be retained? Who can export progress reports?

TEAM / REVIEW QUESTIONS

What must pass in the final demo? Which artifact types matter most? Who may approve high-risk operations?

05 / BUSINESS SCOPE

Problem, outcomes and boundaries

Control scope around one compelling end-to-end learning and project-delivery journey.

PROBLEM AND INTENDED VALUE

Learners may know how to build and evaluate an AI model but lack the engineering judgement needed to deliver a secure, testable, deployable product. Model2Product AI should teach the decisions, not hide them behind automatic code generation.

OUTCOME REQUIREMENTS

B-01	Guide learners through a complete decision-led engineering workflow.
B-02	Require explicit rationale and understanding before high-impact implementation.
B-03	Produce reviewable technical artifacts and a demonstrable working web app.
B-04	Expose project status and learner evidence to instructors / reviewers.

IN SCOPE - GRADUATION MVP

CORE EXPERIENCE

AI web-application projects, staged journey, option comparisons, knowledge-grounded guidance and saved decisions.

IMPLEMENTATION EVIDENCE

Two or more artifacts, isolated code scaffold and test, plus a working deployed web application.

OUT OF SCOPE UNLESS SEPARATELY APPROVED

- Training a foundation LLM, full autonomous product generation or unrestricted repository/cloud changes.
- Deep IDE integration, enterprise organization dashboards and cross-domain learning pathways.
- Support for every proposed specialist agent or a production-level commercial SaaS rollout.

SCOPE NUANCE

Proposal vision suggests 5-6 core agents; minimum demo acceptance requires at least 3 specialist agents sharing state. Keep these separate.

06 / USER EXPERIENCE

The guided project journey

A state machine - not an open-ended chat - determines the active stage, gates and next action.

PROPOSED INTERFACE JOURNEY

01
Onboard
Skill profile + goals

>

02
Discover
Clarify problem / user

>

03
Define
Requirements + MVP

04
Design
Architecture / data

>

05
Integrate
AI, APIs and frontend

>

06
Validate
Safety, testing, release

STAGE-LEVEL INPUT / DECISION / OUTPUT CONTRACT

STEP	EXPECTED SYSTEM BEHAVIOR
Entry	Prerequisites and missing information are checked before the stage opens.
Learn	System explains the task, relevant concepts and realistic alternatives.
Decide	Learner selects and explains a high-impact design choice.
Verify	Rubric checks understanding; weak or incomplete reasoning gets actionable feedback.
Approve	Approved decision is logged before artifact / code actions are unlocked.
Advance	QA checks artifacts, saves state and enables the next eligible stage.

CONTROL PRINCIPLE

A learner requesting implementation too early must receive stage guidance, not an unauthorized code-generation shortcut.

07 / FUNCTIONAL REQUIREMENTS

Discovery, choices and memory

Original FR identifiers preserved from the proposal. Priority tags are a suggested MVP interpretation.

FUNCTIONAL REQUIREMENTS CATALOGUE

FR-01	**User profile and baseline skill assessment** Learner can provide experience, preferences and starting capability.	MUST
FR-02	**Natural-language project discovery conversation** Mentor asks targeted questions to define the problem and expected product.	MUST
FR-03	**Project file upload and context extraction** Proposal / notes can inform discovery while remaining untrusted content.	MUST
FR-04	**Tech Lead-controlled stage sequencing** Workflow service selects the next eligible engineering stage.	MUST
FR-05	**Stage locks and completion criteria** System blocks completion or premature progression until gates pass.	MUST
FR-06	**Structured comparison of design alternatives** Each choice presents pros, cons, constraints and recommended fit.	MUST
FR-07	**Justification and understanding checkpoints** User defends decisions; rubric-based feedback records gaps.	MUST
FR-08	**Persistent project memory and decision log** Choices, constraints and project artifacts survive new sessions.	MUST

TRACKING CONVENTION

The proposal defines 22 functional requirement IDs (FR-01 through FR-22). Keep IDs stable in issues, commits, tests and acceptance evidence.

08 / FUNCTIONAL REQUIREMENTS

Agents and implementation

Original FR identifiers preserved from the proposal. Priority tags are a suggested MVP interpretation.

FUNCTIONAL REQUIREMENTS CATALOGUE

FR-09	**Diagrams and technical artifact generation** System generates structured artifacts from approved decisions.	MUST
FR-10	**Specialist-agent consultation and review** Specialists share context and cross-check outputs.	MUST
FR-11	**RAG over curated technical knowledge** Recommendations use relevant technical evidence and source context.	MUST
FR-12	**Implementation assistance after approval** User enters controlled learn, pair, assist, review or recovery mode.	MUST
FR-13	**Scaffold generation in an isolated workspace** Selected files can be created/modified and automatically checked safely.	MUST
FR-14	**Project tasks and role allocation** System suggests engineering tasks, dependencies and ownership.	SHOULD
FR-15	**Testing and security checklist generation** System produces practical checks and alerts for missing safeguards.	SHOULD
FR-16	**Deployment readiness planning** System covers Docker, secrets, logging, monitoring and release checks.	SHOULD

TRACKING CONVENTION

The proposal defines 22 functional requirement IDs (FR-01 through FR-22). Keep IDs stable in issues, commits, tests and acceptance evidence.

09 / FUNCTIONAL REQUIREMENTS

Progress and delivery outputs

Original FR identifiers preserved from the proposal. Priority tags are a suggested MVP interpretation.

FUNCTIONAL REQUIREMENTS CATALOGUE

FR-17	**Progress dashboard and stage map** User sees completed, active and pending stages and blockers.	MUST
FR-18	**Gamification (XP, badges, milestones)** Rewards follow demonstrated engineering understanding and completion.	SHOULD
FR-19	**Learner competency reporting** System exposes strength, weak areas and evidence behind scores.	MUST
FR-20	**Instructor progress and contribution view** Instructor sees learner decisions, progress and AI contribution.	SHOULD
FR-21	**Exportable artifacts and documentation** User can export a coherent approved technical document package.	MUST
FR-22	**GitHub integration or export-ready repository** User obtains a reviewable GitHub project or export-ready structure.	MUST

PRIORITY INTERPRETATION - TO CONFIRM

MUST VS. SHOULD

MUST items cover the minimum complete demo; SHOULD items improve product quality and teaching value. The supervisor must validate any moved / deferred requirement.

10 / USER STORIES

Prioritized learner stories

Analyst-formulated stories based on the proposal; final wording and priority require stakeholder review.

US-01	As a learner I want to create a project and state my skill level, so the mentor can tailor guidance.	MUST	FR-01, 02
US-02	As a learner I want to upload an idea / proposal, so the mentor understands my context.	MUST	FR-03
US-03	As a learner I want to see the current stage and its exit conditions, so I know what must be done next.	MUST	FR-04, 05, 17
US-04	As a learner I want alternatives compared with trade-offs, so I can choose based on constraints.	MUST	FR-06, 11
US-05	As a learner I want to justify my decision and receive a challenge, so I learn the reasoning, not just an answer.	MUST	FR-07
US-06	As a learner I want previous decisions restored when I return, so I can continue without repeating context.	MUST	FR-08
US-07	As a learner I want approved diagrams or API specs generated, so decisions produce tangible artifacts.	MUST	FR-09

STORY READINESS CHECKLIST

Each story needs a named user, business outcome, linked FR IDs and a testable pass/fail condition before entering development.

11 / USER STORIES

Delivery and review stories

Analyst-formulated stories based on the proposal; final wording and priority require stakeholder review.

US-08	As a learner I want implementation tools available after approval, so I can build safely with support.	MUST	FR-12, 13
US-09	As a learner I want to see tasks, release checks and risks, so I do not miss engineering work.	SHOULD	FR-14, 15, 16
US-10	As a learner I want a competency / progress dashboard, so I can see what I actually learned.	MUST	FR-17, 19
US-11	As an instructor I want to review decisions and AI contribution, so I can evaluate real learner ownership.	SHOULD	FR-20
US-12	As a learner I want to export artifacts and project files, so I can submit and demonstrate my work.	MUST	FR-21, 22
US-13	As a learner I want milestones and skill-focused rewards, so progress remains engaging without rewarding random clicks.	SHOULD	FR-18
US-14	As a reviewer I want a readiness and evidence summary, so I can verify acceptance criteria efficiently.	MUST	FR-15, 16, 19

STORY READINESS CHECKLIST

Each story needs a named user, business outcome, linked FR IDs and a testable pass/fail condition before entering development.

12 / USE CASES

UC-01 / UC-02 : guide and decide

Use-case preconditions and alternative paths are draft acceptance interpretations.

UC-01

Project discovery

ACTOR	Learner
PRECONDITION	Authenticated project owner with a new or existing project.
MAIN SUCCESS FLOW	Enter project idea -> optionally upload proposal -> system identifies missing context -> learner answers -> brief and scope are generated -> learner approves.
ALTERNATE FLOW	If facts are missing, system asks focused follow-ups; uploaded instructions do not override policy.
POSTCONDITION	Approved brief and scoped requirements are saved to the project.

UC-02

Engineering decision checkpoint

ACTOR	Learner / Tech Lead Orchestrator
PRECONDITION	Current stage is unlocked and its decision context is complete.
MAIN SUCCESS FLOW	Agent retrieves evidence -> presents 2+ options and trade-offs -> learner selects / explains -> rubric evaluates -> approval is stored -> next allowed action unlocks.
ALTERNATE FLOW	Weak justification triggers feedback / retry; early build requests remain blocked.
POSTCONDITION	Decision, explanation, evaluation and stage state are persistently recorded.

13 / USE CASES

UC-03 / UC-04 : build and review

Controlled tool execution and evidence export make the journey more than conversational guidance.

UC-03

Generate and validate an artifact

ACTOR	Learner / specialist agent
PRECONDITION	Required architectural choice has been approved.
MAIN SUCCESS FLOW	Choose artifact -> agent creates structured output -> schema / QA checks run -> user reviews result -> accepted revision is saved with source decision.
ALTERNATE FLOW	If checks fail, show errors and offer revision; do not replace prior approved artifact silently.
POSTCONDITION	Versioned artifact appears in project workspace and export package.

UC-04

Implementation and progress review

ACTOR	Learner / instructor
PRECONDITION	Project state exists; tool access follows role and stage permissions.
MAIN SUCCESS FLOW	Authorize selected scaffold -> run isolated check/test -> capture output -> view stage progress and competency evidence -> optionally export report.
ALTERNATE FLOW	Unsafe or destructive operations require explicit approval; failed tool operations do not advance the stage.
POSTCONDITION	Tool audit, test result and progress report remain linked to project history.

14 / QUALITY ATTRIBUTES

Non-functional requirements

The proposal specifies quality categories; exact numeric service levels remain to be defined and approved.

NFR-01	**Usability**	Show current stage, its purpose, required actions and next step.
NFR-02	**Reliability**	Persist decisions and resume state after failure / session restart.
NFR-03	**Performance**	Provide visible progress for long tool calls; keep mentor interaction responsive.
NFR-04	**Security**	Authenticate roles, validate inputs, sandbox execution, protect secrets and log critical actions.
NFR-05	**Privacy**	Isolate user/project data and offer retention / deletion controls.
NFR-06	**Explainability**	Show decision rationale, alternatives and retrieved-source context.
NFR-07	**Maintainability**	Use modular agents/tools, documented workflows, typed APIs and automated tests.
NFR-08	**Scalability**	Support isolated stores and a path to queueing long-running work.
NFR-09	**Observability**	Capture tool outcomes, latency, token / cost, retrieval results and errors.
NFR-10	**Accessibility**	Support keyboard use, readable contrast, clear labels and responsive layouts.

VERIFICATION APPROACH

Review UI flows, reload persistence, execute security/permission tests, inspect observability traces and run accessible UI checks. Define numeric targets during validation.

15 / DATA AND AI

State, knowledge and permission needs

Stable structured project state must remain authoritative; LLM outputs should not bypass workflow rules.

PROPOSED CORE LOGICAL DATA OBJECTS

ENTITY GROUP	DATA REQUIRED / INTENDED USE
User / Project	User identity, skill profile, title, problem, target users, status and owner.
Stage / Decision	Stage gates, option set, selection, written justification and review result.
Artifact / Task	Type, version, origin, approval, storage path; assignee and dependency.
AgentRun / AuditEvent	Agent/tool actions, output, latency, tokens, errors and approval events.
KnowledgeChunk	Approved technical content, embedding, version, source and review date.
Competency	Skill evidence, evaluation signal, confidence and update timestamp.

AI-SYSTEM BEHAVIORAL CONSTRAINTS

- Tech Lead orchestrator alone controls stage transitions and tool authorization.
- Specialist agents share the same approved project context and structured output schemas.
- RAG answers should cite relevant knowledge; gaps in evidence should be disclosed.
- Decisions, rubrics and artifacts must be auditable; uploads and retrieved text are untrusted.
- Generated code must be limited to isolated workspace operations with human approval gates.

INTEGRATION DIRECTION - PROPOSED, NOT MANDATORY

Proposal recommends a React/Next.js UI, FastAPI services, PostgreSQL, pgvector / Qdrant, a state-machine / LangGraph coordinator and isolated Docker-based tools.

16 / TRACEABILITY

Minimum MVP acceptance matrix

These ten checks are retained from the proposal and linked to requirement IDs and observable evidence.

AC-01	Create an AI project and complete 4 guided stages.	FR-01,02,04,05,17 Completed journey / project log
AC-02	Block unauthorized premature implementation.	FR-04,05,07,12 Negative test / stage denial
AC-03	Run at least 3 specialists over shared state.	FR-08,10 Agent traces + shared decision
AC-04	Retrieve evidence from vector knowledge base.	FR-11 Query + cited retrieval record
AC-05	Provide 2 option comparisons with justification.	FR-06,07 Two saved choice / rationale records
AC-06	Generate 2 artifacts from approved decisions.	FR-09 Diagram + ERD or API specification
AC-07	Generate / modify a small scaffold and test it.	FR-12,13 Sandbox logs + test result
AC-08	Persist decisions and artifacts between sessions.	FR-08,09 Sign out/in continuity test
AC-09	Show progress and competency indicators.	FR-17,19 Dashboard inspection
AC-10	Deploy web app with tests, GitHub and README.	FR-13,15,16,22 URL + repo + tests + README

ACCEPTANCE RULE

All minimum checks require repeatable evidence; demo appearance alone is insufficient. The supervisor approves the final acceptance thresholds.

17 / UNRESOLVED DECISIONS

Open questions and assumption log

Requirements gathering is incomplete until these questions have named owners and signed decisions.

Q-01	Who may create, view or grade projects? Are instructor roles required in the first demo?
Q-02	Which upload formats, limits and file-retention policy are acceptable?
Q-03	How does the rubric score a justification, and what counts as passing?
Q-04	What is the precise stage order, and where are branches / retries allowed?
Q-05	Which 3+ specialist agents and 4 stages must work end to end?
Q-06	Which artifacts are mandatory and what is the approval / versioning flow?
Q-07	Which LLM and technical sources are allowed, and what budget / latency target applies?
Q-08	Which test, performance, privacy and accessibility thresholds are feasible?

INITIAL ASSUMPTION REGISTER

A-01: 4 members will share delivery ownership. A-02: graduation MVP focuses on AI-enabled web apps. A-03: priorities in this document are proposals, not stakeholder-approved commitments.

DECISION LOG FORMAT

Record: question ID | owner | alternatives | decision | justification | date | affected FR / NFR / story IDs.

18 / REVIEW AND APPROVAL

Validation checklist & references

The document becomes a signed baseline only after stakeholder priorities, scope and acceptance criteria are agreed.

PROPOSED FOUR-MEMBER OWNERSHIP FOR REQUIREMENTS

ROLE	REQUIREMENTS RESPONSIBILITY
M1 - Product / Frontend / UX	Stakeholder sessions, scope, UI flows, issue tracking and sign-off.
M2 - Agentic AI / Orchestration	Agent flows, stage gates, persistent memory and AI behavioral requirements.
M3 - RAG / Knowledge / Evaluation	RAG sources, retrieval quality, rubric design and evaluation evidence.
M4 - Backend / DevOps / QA	Data, APIs, security, testing, CI/CD and deployment constraints.

SIGN-OFF CHECKLIST (COMPLETE AFTER ELICITATION)

- [ ] Stakeholders and user personas validated with real participants.
- [ ] Scope, Must/Should priority labels and deferrals approved.
- [ ] FRs, NFRs and acceptance evidence reviewed for testability.
- [ ] Permissions, uploads, rubric thresholds and retention policy resolved.
- [ ] Traceability and changes logged in version-controlled backlog.

APPROVALS

Team / product owner Name: _____ Date: _____

Instructor / supervisor Name: _____ Date: _____

SOURCE AND ENGINEERING REFERENCES

Internal: Model2Product AI Graduation Project Proposal (Aug 2026)
Optional: OWASP ASVS
Optional: W3C WCAG 2.2